$$((x-1)+1)^2 = (x-1)^2 + 2(x-1) + 1$$
$$(x-1)^2 = (x-2)^2 + 2(x-2) + 1$$
$$x^2 = \sum_{b=0}^{x-1} [2b+1]$$
$$((x-1)+1)^1 = (x-1)^1 + 1 \; ((x-1))^1 = (x-2)^1 + 1$$
$$x^1 = \sum_{b=0}^{x-1} [1]$$
$$x^2 = \sum_{b=0}^{x-1} \left[2 \sum_{c=0}^{b-1} [1] + 1\right]$$
$$x^2 = 2 \sum_{b=0}^{x-1} \left[\sum_{c=0}^{b-1} [1]\right] + 1 \sum_{b=0}^{x-1} [1]$$

I have a theory, all x^n's can be written as a sum of nested sums with falling upper bounds

$$x^n = d_1 \sum_{b=0}^{x-1} 1 + d_2 \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} 1 + ... + d_n \overbrace{\sum_{b=0}^{x-1} \sum_{c=0}^{b-1} ... \sum_{n_n=0}^{n_{n-1}-1}}^{n \text{ sums}} 1$$

This the goal of this paper

Reucrsions from Binomial therom

falling sums

according to formulas.typ's falling_sum section.

for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid d$ in $\mathbb{R}$ where $\left(\frac{x}{d} = l, l \in \mathbb{Z}^+\right) \Rightarrow$

$$x^r = \overbrace{\sum_{b=0}^{r-1} \left[\binom{r}{b} a^{r-b} \left[\sum_{c=0}^{b-1} \left[\binom{b}{c} a^{b-c} ... \left[\sum_{n_l=0}^{n_{l-1}-1} \left[\binom{n_{l-1}}{n_l} a^{n_{l-1}-n_l} (x-la)^{n_l} \right] \right] ... \right] \right] \right]}^{l \text{ sums}}$$
$$+ \overbrace{\sum_{b=0}^{r-1} \left[\binom{r}{b} a^{r-b} \left[\sum_{c=0}^{b-1} \left[\binom{b}{c} a^{b-c} ... \left[\sum_{n_{l-1}=0}^{n_{l-2}-1} \left[\binom{n_{l-2}}{n_{l-1}} a^{n_{l-2}-n_{l-1}} (x-la)^{n_{l-1}} \right] \right] ... \right] \right] \right]}^{l-1 \text{ sums}}$$
$$+ ... + \overbrace{\sum_{b=0}^{r-1} \left[\binom{r}{b} a^{r-b} [(x-2a)^b] \right]}^{1 \text{ sums}}$$
$$+ \overbrace{(x-a)^r}^{0 \text{ sums}}$$

I will extend this paper by attempting to calculate the scalars on the sums (the d_n numbers):

$$x^n = d_1 \sum_{b=0}^{x-1} 1 + d_2 \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} 1 + ... + d_n \overbrace{\sum_{b=0}^{x-1} \sum_{c=0}^{b-1} ... \sum_{n_n=0}^{n_{n-1}-1}}^{n \text{ sums}} 1$$

the example in this paper was derived from the binomial therom, the outer term was expanded to generate recurssion

this is excalty what falling_sum_lr_btherom does (upto 2 sums):

for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$

$$x^r = \sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} (x-(b+1)a)^c a^{r-c} \right] \right] + (x-la)^r$$

lets revist our inital example for x^2

$$\text{let } a = 1, l = x$$

$$\text{for all } 1 \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid x \in \mathbb{Z}^+$$

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} (x - (b+1))^c \right] \right] + (0)^r$$

$$\text{let } r = 2$$

$$x^2 = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{2-1} \left[\binom{2}{c} (x - (b+1))^c \right] \right] + (0)^2$$

$$x^2 = \sum_{b=0}^{x-1} \left[\binom{2}{0} (x - (b+1))^0 + \binom{2}{1} (x - (b+1))^1 \right] + (0)^2$$

$$x^2 = \sum_{b=0}^{x-1} [1 + 2(x - (b+1))]$$

$$x^1 = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{1-1} \left[\binom{1}{c} (x - (b+1))^c \right] \right] + (0)^1$$

$$x^1 = \sum_{b=0}^{x-1} \left[\binom{1}{0} \right]$$

$$x^2 = \sum_{b=0}^{x-1} \left[1 + 2 \left(\sum_{c=0}^{x-1} [1] - (b+1) \right) \right]$$